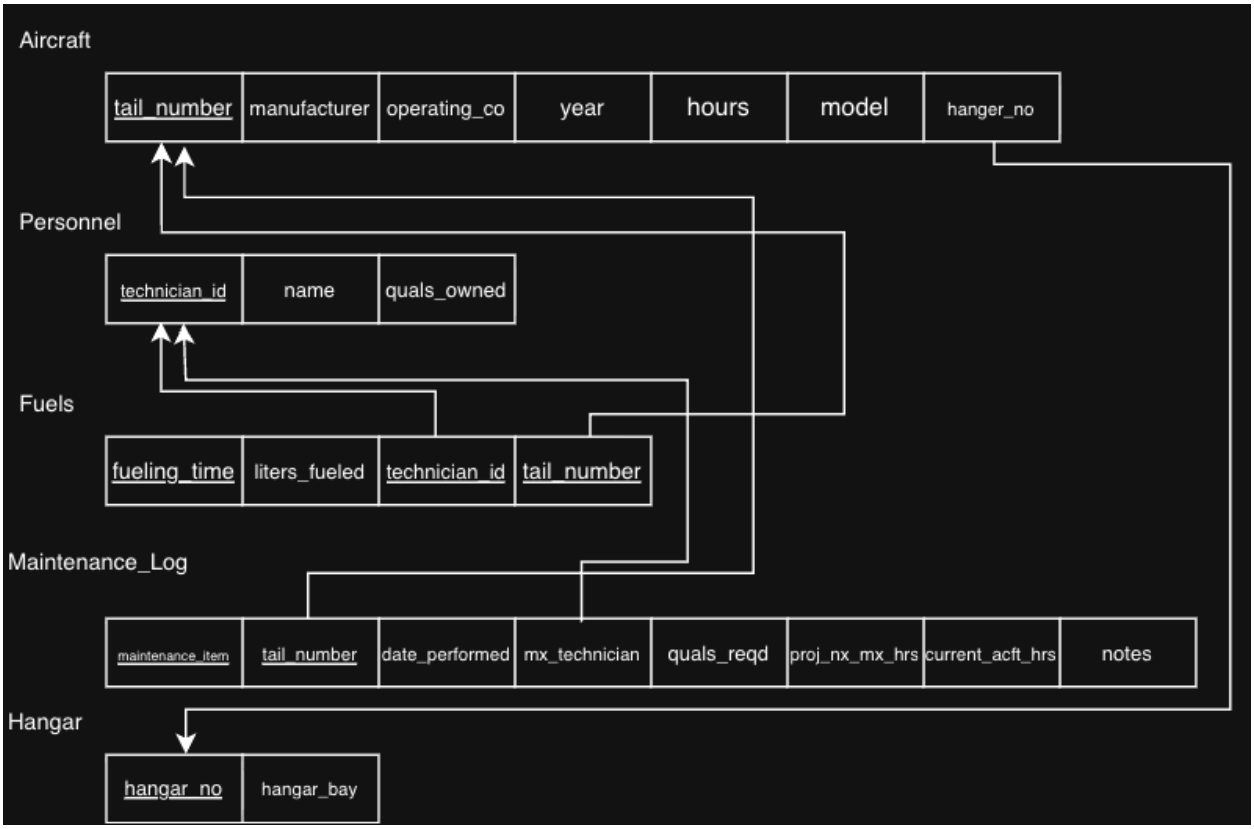
# Group Members:

Erfon, Ben, Brendan, and Chris

# Relational Diagram:




```
+-------------+--------------+--------------+------+---------+--------------------+-----------+
| tail_number | manufacturer | operating_co | year | hours   | model              | hangar_no |
+-------------+--------------+--------------+------+---------+--------------------+-----------+
| 89-0487     | Lockheed     | USAF         | 1989 | 30033.5 | F-15E Strike Eagle |         3 |
| 89-1690     | Lockheed     | USAF         | 1989 | 29033.5 | F-15E Strike Eagle |         2 |
+-------------+--------------+--------------+------+---------+--------------------+-----------+
+-----------+-------------+
| hangar_no | hangar_bay  |
+-----------+-------------+
|         1 | Alpha       |
|         2 | Bravo       |
|         3 | Charlie     |
+-----------+-------------+
+------------------+-------------+---------------------+---------------+------------+------------------+----------------------+--------------------------------+
| maintenance_item | tail_number | date_performed      | mx_technician | quals_reqd | current_acft_hrs | projected_next_mx_hrs | notes                          |
+------------------+-------------+---------------------+---------------+------------+------------------+----------------------+--------------------------------+
| Fuel Line Flush  | 89-1690     | 2026-11-08 21:25:00 | 123456789     | Fuels      |          29033.5 |              32033.5 | Fuel Lines successfully flushed |
+------------------+-------------+---------------------+---------------+------------+------------------+----------------------+--------------------------------+
+---------------+--------------+--------------+
| technician_id | name         | quals_owned  |
+---------------+--------------+--------------+
| 011899988     | M. Mario     | Avionics     |
| 123456789     | G. Dlugiwicz | Fuels        |
+---------------+--------------+--------------+
+---------------------+---------------+---------------+-------------+
| fueling_time        | liters_fueled | technician_id | tail_number |
+---------------------+---------------+---------------+-------------+
| 2026-11-08 22:00:00 |       304.560 | 011899988     | 89-1690     |
+---------------------+---------------+---------------+-------------+
```

# Sample Data input into Tables:

# SQL:

```
drop database if exists project_group;
create database project_group;
use project_group;

drop table if exists hangar;
drop table if exists maintenance_log;
drop table if exists fuels;
drop table if exists personnel;
drop table if exists aircraft;


create table aircraft(
                tail_number varchar(30) primary key,
                manufacturer varchar(30),
                operating_co varchar(30),
                year int,
                hours float(20,1),
                model varchar(50),
                hangar_no int

);


create table hangar (
                hangar_no int primary key,
                hangar_bay varchar(10)
);

create table personnel(
                technician_id varchar(20) primary key,
                name varchar(30),
                quals_owned varchar(40)
);

create table maintenance_log(
                maintenance_item varchar(50) primary key,
                tail_number varchar(30),
                date_performed datetime,
                mx_technician varchar(20),
                quals_reqd varchar(40),
                current_acft_hrs float(20,1),
                projected_next_mx_hrs float(20,1),
                notes text,
                -- constraints
                constraint fk_tn
                        foreign key (tail_number)
                        references aircraft(tail_number)
                                on delete cascade
                                on update cascade,
```

```
                constraint fk_mx
                        foreign key (mx_technician)
                        references personnel(technician_id)
                                on delete cascade
                                on update cascade
);
create table fuels(
        fueling_time datetime primary key,
        liters_fueled float(10,3),
        technician_id varchar(20),
        tail_number varchar(30),
                constraint fk_tn2
                        foreign key (tail_number)
                        references aircraft(tail_number)
                                on delete cascade
                                on update cascade,
                constraint fk_mx2
                        foreign key (technician_id)
                        references personnel(technician_id)
                                on delete cascade
                                on update cascade
);
alter table aircraft
add constraint fk_hangar
foreign key (hangar_no)
references hangar(hangar_no);
-- triggers
delimiter $$
-- calculate the next maintenance hour number that the aircraft will require
create trigger calc_next_mx_1
before insert on maintenance_log
for each row
begin
   if NEW.projected_next_mx_hrs is null then
      set NEW.projected_next_mx_hrs = NEW.current_acft_hrs + 100.0;
   end if;
end$$
delimiter ;

delimiter $$
-- check that the maintenance technician has appropriate qualifications for task
create trigger check_quals_mx
before insert on maintenance_log
for each row
begin
   declare v_owned_quals varchar(40);

   select quals_owned into v_owned_quals
   from personnel
   where technician_id = NEW.mx_technician;

   -- verify tech qual matches the required task qual
   if NEW.quals_reqd is not null and (v_owned_quals is null or v_owned_quals not like
concat('%', NEW.quals_reqd, '%')) then
```

```
        signal sqlstate '45000'
        set message_text = 'Error: Assigned technician does not hold the required quals for this
maintenance item.';
    end if;
end$$
delimiter ;

-- adding in some sample data
insert into hangar(hangar_no, hangar_bay)
values
                (1, "Alpha"),
                (2, "Bravo"),
                (3, "Charlie");
insert into aircraft(tail_number, manufacturer, operating_co, year, hours, model, hangar_no)
values
                ("89-1690", "Lockheed", "USAF", 1989, 29033.5, "F-15E Strike Eagle", 2),
                ("89-0487", "Lockheed", "USAF", 1989, 30033.5, "F-15E Strike Eagle", 3);

insert into personnel(technician_id, name, quals_owned)
values
                ("123456789", "G. Dlugiwicz", "Fuels"),
                ("011899988", "M. Mario", "Avionics");

insert into maintenance_log(maintenance_item, tail_number, date_performed, mx_technician,
quals_reqd, current_acft_hrs, projected_next_mx_hrs, notes)
values
                ("Fuel Line Flush", "89-1690", "2026-11-08 21:25:00", "123456789", "Fuels",
29033.5, 32033.5, "Fuel Lines successfully flushed");
insert into fuels(fueling_time, tail_number, liters_fueled, technician_id)
values
                ("2026-11-08 22:00:00", "89-1690", 304.56, "011899988");

-- display sample data
select * from aircraft;
select * from hangar;
select * from maintenance_log;
select * from personnel;
select * from fuels;
```